

Dr. Eric Janusson

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Professional Summary

PhD Analytical Chemist and Mass Spectrometrists specializing in analytical method development, with deep expertise in high-resolution mass spectrometry, analytical instrumentation, and scientific computing. Experienced in developing and validating complex analytical workflows, troubleshooting and maintaining analytical instrumentation, and developing computational tools for data analysis and automation. Proven track record of commissioning analytical facilities, implementing ISO/IEC 17025-aligned operations, leading capital procurement, and delivering analytical solutions in industrial environments.

Core Competencies

- **Mass Spectrometry & Analytical Instrumentation:** LC-MS/MS, GC-MS, TWIMS, IC; QTOF, QqQ, Orbitrap, and magnetic-sector analyzers; ESI, ICP, EI, and custom ion sources; instrument commissioning, troubleshooting, and maintenance.
- **Scientific Computing & Analytical Informatics:** Python, pandas, matchms, programmatic data analysis, database architecture, automated analytical workflows, and scientific software development.
- **Method Development & Laboratory Systems:** Analytical method development and validation, trace analysis, analytical workflow design, laboratory commissioning, and capital equipment procurement.
- **Regulated R&D & Quality Systems:** ISO/IEC 17025 and cGMP-aligned laboratory operations, 5N+ purity certification, Health Canada Section 56 compliance, SOP development, and chain-of-custody controls.

Professional Experience

Principal Scientist, Analytical Development & Laboratory Operations | TMC Manufacturing | North Vancouver, BC | 2024 – 2026

- Performed daily chemical analysis to optimize chemical purification and engineering processes for ultra-high-purity ($\geq 99.999\%$, 5N+) rare-earth and inorganic stable isotopes.
- Developed and validated a bespoke potentiometric assay with 1 ppm detection limit, reducing halogen contamination by 99.997% and effectively eliminating out-of-specification batches.
- Led \$515K CAD in capital procurement for automated ICP-MS and IC instrumentation, establishing an in-house laboratory framework for purity verification.
- Designed and validated trace-element analytical methods replacing external testing with an automated ISO/IEC 17025-aligned platform.
- Managed industrial-scale ultra-high-purity stable isotope production, including operation and maintenance of enrichment systems, custom ion sources, electromagnetic ion optics, UHV systems, gas manifolds, and high-voltage electronics.
- Supervised and mentored cross-functional chemistry and engineering staff, directly contributing to the promotion of two junior scientists.

Lead Chemist | Delic Labs | Vancouver, BC | 2021 – 2022

- Commissioned a federally authorized analytical facility from the ground up, successfully installing, qualifying, and maintaining high-resolution LC-MS, GC-MS, HPLC, and spectroscopic (FTIR, UV-Vis) platforms.
- Authored Standard Operating Procedures (SOPs) and defined an analytical services framework strictly aligned with ISO/IEC 17025.
- Secured Health Canada Section 56 exemptions for Schedule I controlled substances, implementing strict, audit-ready inventory and chain-of-custody controls.

Postdoctoral Research Associate (Cronin Group) | University of Glasgow | Glasgow, UK | 2017 – 2020

- Designed multidimensional LC-TWIMS-MS analytical methods, integrated with a robotic synthetic platform, to resolve the real-time self-assembly mechanisms of inorganic polyoxometalate (POM) nanomaterials.
- Discovered and published a novel carbohydrate-directed synthetic route conferring isomer and conformational control over self-assembling POM clusters.

PhD Researcher & Graduate Laboratory Instructor | University of Victoria | Victoria, BC | 2012 – 2017

- Developed real-time ESI-MS/MS (QTOF, QqQ) methodologies to detect, characterize, and monitor transient organometallic intermediates natively in solution.
- Published foundational studies on electrospray ionization (ESI) ion-source physics, ion pairing, solvent effects, and charge-tagging strategies for selective mass spectrometric analysis.
- Supervised senior undergraduates through independent, publication-level laboratory research.

Education

- **PhD, Analytical Chemistry** | University of Victoria, BC (2017)
 - *Dissertation*: "Development of mass spectrometry techniques for real-time reaction monitoring."
- **BSc, Chemistry** | University of Victoria, BC (2012)

Certifications

- Agilent LC-QTOF, GC-MS, FTIR, and UV-Vis Operator Training & Certification (2021)
- Waters Synapt G2 TWIMS & QTOF Operator Training & Certification (2018)

Selected Publications

- Janusson, E. et al. "Synthesis of polyoxometalate clusters using carbohydrates as reducing agents leads to isomer-selection." *Chem. Commun.*, 2019.
- Janusson, E. et al. "Step-by-step real-time monitoring of a catalytic amination reaction." *Chem. Commun.*, 2019.
- Janusson, E. et al. "Real-time analysis of Pd₂(dba)₃ activation by phosphine ligands." *Chem. Commun.*, 2017.

Full list of publications available at <https://orcid.org/0000-0002-3207-7067>